If X has more than two points and (X, τ) be an indiscrete topology then which of the following statement is true about (X, τ) ?

Select the correct option

Reload Math Equations

○	It is not metrizable.
○	It is Haussdorff.
○	None of them.
○	It is metrizable.

Question # 5 of 10 (Start time: 08:44:28 PM, 30 December 2024) **Total Marks:**

A topological space is said to be the second countable space if it has a .....base.

Select the correct option

Reload Math Equations

- uncountable open
- uncountable closed
- countable open
- countable closed

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 08:47:09 PM, 30 December 2024) **Total Marks:** 1

Which of the following best describes a projection map in topology?

Select the correct option

Reload Math Equations

- ○ A map that assigns every point in a Cartesian product space to its first component.
- ○ A continuous map from a topological space to a discrete set.
- ○ A homeomorphism between two topological spaces.
- ○ A map that preserves the open sets of a topological space.

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 09:17:24 PM, 30 December 2024) **Total Marks:** 1

Let (X, τ) be a metrizable then which of the following statement is true?

Select the correct option

Reload Math Equations

○	(X, τ) is second countable.
○	(X, τ) has the countable chain collection
○	(X, τ) is separable.
○	All of them

Click to Save Answer & Move to Next Question

9:22 PM

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BC230430302: MEHBOOB AHMAD

MTH634 - Topology (Quiz No. 3) Quiz Start Time: 09:22 PM

Question # 1 of 10 (Start time: 09:22:44 PM, 30 December 2024)

If (X,d) is a compact metric space, then_____

Select the correct option

Reload

- every sequence which has a convergent subsequence is a Cauchy sequence.
- every sequence has a subsequence which is a Cauchy sequence.
- every Cauchy sequence converges.
- every sequence is a Cauchy sequence.

Click to Save Answer & Move

9:32

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MTH634 - Topology (Quiz No. 3)

Question # 7 of 10 (Start time: 09:31:50 PM, 30 December 2024)

Which of the following statement is true?

Select the correct option

- Continuous map may or may not be an open map
- Continuous map is always an open map.
- Continuous map is always a closed map.
- none of these

Click

A map 'f' from a topological space X to a topological space Y is .....if and only if the image of each closed subset of X is closed in Y.

elect the correct option

- a surjective map
- a closed map
- an injective map
- an open map

Click to Save Answer & Move to Next Questio

Question # 3 of 10 (Start time: 09:36:23 PM, 30 December 2024) **Total Marks:** 1

$$T_{X \times Y}$$

denotes .....on $X \times Y$.

Select the correct option

Reload Math Equations

- product topology
- co finite topology
- lower limit topology
- upper limit topology

Click to Save Answer & Move to Next Question

9:36 PM

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BC230430302: MEHBOOB AHMAD

MTH634 - Topology (Quiz No. 3)

Question # 8 of 10 (Start time: 09:36:41 PM, 30 December 2024)

A topological space (X,T) is said to be metrizable if there is a metric d on X that generate... ...

Select the correct option

- ○ none of these
- ○ topology T
- ○ whole space X
- ○ both X and T



Question # 7 of 10 (Start time: 09:38:14 PM, 30 December 2024) **Total Marks:**

Let X and Y be topological spaces. A map $f : X \to Y$ is called a Closed Map if_____

Select the correct option

Reload Math Equations

○	for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed.
○	None of them
○	for every closed set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed
○	for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is open.

Click to Save Answer & Move to Next Question

10° Search

BC230430302: MEHBOOB AHMAD

MTH634 - Topology (Quiz No. 3) Quiz Start Time: 09:22 P

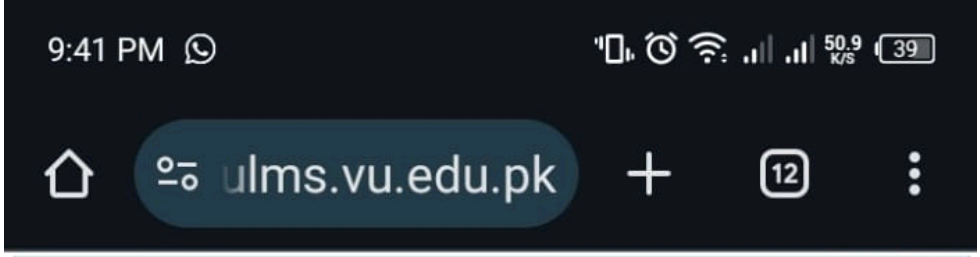
Question # 9 of 10 (Start time: 09:38:04 PM, 30 December 2024)

A map 'f' between two topological spaces is called ..... iff 'f' is bijective, continuous and f-1(f inverse) is continuous.

Select the correct option

- isomorphism
- automorphism
- homeomorphism
- bijection

Click to Save Answer & Mo



BC230211223: TANZILA BIBI

Time Left 86 soc(s)

MTH634 - Topology (Quiz No. 3)

Quiz Start Time: 09:41 PM, 30 December 2024

Question # 1 of 10 (Start time: 09:41:05 PM, 30 December 2024)

Total Marks: 1

Every metric space is first countable.

Select the correct option

Reload Math Equations

- True
- False

Click to Save Answer & Move to Next Question



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BC230211223: TANZILA BIBI

Time Left 89 soc(s)

MTH634 - Topology (Quiz No. 3)

Quiz Start Time: 09:41 PM, 30 December 2024

Question # 2 of 10 (Start time: 09:41:38 PM, 30 December 2024)

Total Marks: 1

Let X and Y be topological spaces. A map $f : X \to Y$ is called an Open Map if_____

Select the correct option

Reload Math Equations

- for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed.
- for every closed set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed.
- None of them
- for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is open.

Click to Save Answer & Move to Next Question

9:45 PM

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BC230211223: TANZILA BIBI

MTH634 - Topology (Quiz No. 3) Quiz Start

Question # 8 of 10 (Start time: 09:45:27 PM, 30 December 2024)

Which of the following statements best defines a metric space?

Select the correct option

- ○ A set that is equipped with a topology.
- ○ A set where every element has a unique inverse element.
- ○ A set with a binary operation that combines any two elements to form a third element.
- ○ A set with a function that measures the distance between any two elements in the set.

Click to Save

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MTH634 - Topology (Quiz No. 3)

Question # 9 of 10 (Start time: 09:46:18 PM, 30 December 2024)

Which of the following statements about open balls in a metric space is true?

Select the correct option

- An open ball always includes its boundary points.
- Every open ball is an open set in the metric space.
- The radius of an open ball can be negative.
- The centre of an open ball is not necessarily contained in the ball.

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MTH634 - Topology (Quiz No. 3) Quiz

Question # 1 of 10 (Start time: 09:51:23 PM, 30 December 2024)

A homoemorphic map between two topological spaces is always .....

Select the correct option

- discontinuous
- surjective but not injective
- bijective
- injective but not surjective

Click to

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Question # 2 of 10 (Start time: 09:52:19 PM, 30 December 2024)

A homoemorphic map between two topological spaces is always .....

Select the correct option

- continuous
- an identity map
- discontinuous
- piecewise continuous

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Question # 3 of 10 (Start time: 09:52:56 PM, 30 December 2024)

Let X and Y be two topological spaces. If f is a function from X to Y, then f is said to be homeomorphism if.....

Select the correct option

- all of these options
- f is bijective
- f is continuous
- f is open

Click to Save Answer &

9:53 PM

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MTH634 - Topology (Quiz No. 3) Quiz Start Ti

Question # 4 of 10 (Start time: 09:53:39 PM, 30 December 2024)

Metric topology induced by $d(x, y) = |x - y|$ on $\mathbb{R}$ is called_____

Select the correct option

- usual topology
- indiscrete topology
- None of them
- discrete topology

Click to Save An

9:55 PM

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BC230406570: UROOJ FATIMA

MTH634 - Topology (Quiz No. 3) Quiz

Question # 6 of 10 (Start time: 09:55:04 PM, 30 December 2024)

Which of the following properties must a relation R on a set A satisfy to be an equivalence relation?

Select the correct option

- antisymmetry and transitivity
- reflexivity, symmetry, and transitivity
- symmetry and transitivity
- reflexivity and symmetry

Click to

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Question # 7 of 10 (Start time: 09:55:46 PM, 30 December 2024)

Let $X = \{1, 2, 3, 4\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X, then which of the following is NOT true ?

Select the correct option

○	The local base of the element 4 is $\emptyset$.
○	(X, τ) be a topological space.
○	(X, τ) be a first countable space.
○	Every element of X has uncountable local base.

Click to Save Answer & N

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Question # 8 of 10 (Start time: 09:56:22 PM, 30 December 2024)

Two spaces are called topologically equivalent if there exists a ..... between two topological spaces.

Select the correct option

- an injective map
- homeomorphism
- surjective map
- bijection

Click to Save Answer & N

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Question # 10 of 10 (Start time: 09:57:49 PM, 30 December 2024)

What is an open cover of a topological space X?

Select the correct option

- A collection of open sets whose union is X.
- A collection of open sets that are disjoint.
- A single open set that contains X.
- A collection of closed sets whose intersection is X.

Click to

Question # 1 of 10 (Start time: 10:18:03 PM, 30 December 2024)

Which of the following statement is true?

Select the correct option

○	Restriction of an open map can never be an open map.
○	Restriction of an open map may or may not be an open map.
○	Restriction of an open map is always an open map.
○	Restriction of a closed map may or may not be an open map.

Question # 2 of 10 (Start time: 10:19:08 PM, 30 December 2024)

A topological property is a property of a topological space that is preserved under homeomorphisms, which of the following represent the topological property .....

Select the correct option

○	All of these options
○	Discreteness
○	Countability
○	Connectedness and Compactness

Question # 3 of 10 (Start time: 10:20:43 PM, 30 December 2024)

Let $X = R$(Set of real numbers) be a usual metric space and $N \subseteq R$, then which of the following is true about N ?

Select the correct option

○	It is not an open set.
○	It has disjoint intersection with the set of its interior points.
○	All of them.
○	None of its point is an interior point.

Question # 4 of 10 (Start time: 10:21:47 PM, 30 December 2024)

If X has more than two points and (X, τ) be an indiscrete topology then which of the following statement is true about (X, τ) ?

Select the correct option

○	None of them.
○	It is not metrizable.
○	It is Haussdorff.
○	It is metrizable.

Question # 6 of 10 (Start time: 10:23:53 PM, 30 December 2024)

A map 'f' from a topological space X to a topological space Y is .....if and only if the image of each closed subset of X is closed in Y.

Select the correct option

- ◯ a closed map
- ◯ an open map
- ◯ an injective map
- ◯ a surjective map

Question # 7 of 10 (Start time: 10:24:22 PM, 30 December 2024)

Which of the following statements about open balls in a metric space is true?

Select the correct option

○	The radius of an open ball can be negative.
○	Every open ball is an open set in the metric space.
○	The centre of an open ball is not necessarily contained in the ball.
○	An open ball always includes its boundary points.

MTH634 - Topology (Quiz No. 3)

Question # 9 of 10 (Start time: 10:26:29 PM, 30 December 2024)

Let $X = R$(Set of real numbers) be a usual metric space and $N \subseteq R$, then which of the following is NOT true about N ?

Select the correct option

○	It is not a neighbourhood of any of its point.
○	None of them.
○	Neither of its point is an interior point.
○	It must be an open set.

Question # 10 of 10 (Start time: 10:27:10 PM, 30 December 2024)

Two spaces are called topologically equivalent if there exists a ..... between two topological spaces.

Select the correct option

○	an injective map
○	bijection
○	surjective map
○	homeomorphism

